

Installation and Execution of Java Servlet Programs

Software Required

You need the following software:

1. Java Development Kit (JDK)
2. Eclipse IDE for Enterprise Java and Web Developers
3. Apache Tomcat Server

Step 1: Install Java JDK

1. Download JDK

Download and install the latest JDK (or the version required by your course).

2. Install JDK

Run the installer and complete the installation.

Example installation path:

C:\Program Files\Java\jdk-21

3. Set JAVA_HOME (Optional but Recommended)

1. Open **Environment Variables**.
2. Create a new variable:

Variable Name : JAVA_HOME

Variable Value :

C:\Program Files\Java\jdk-21

3. Add the JDK's **bin** folder to the **Path** environment variable.

4. Verify Installation

Open **Command Prompt**.

Type:

java -version

Example output:

java version "21.0.2"

Now type:

javac -version

Example:

javac 21.0.2

If both commands work, Java is installed correctly.

Step 2: Install Apache Tomcat

1. Download Tomcat

Download the ZIP version of Apache Tomcat.

2. Extract

Extract it to a folder.

Example:

C:\Apache\Tomcat10

Folder Structure

Tomcat

1. bin
2. conf
3. lib
4. logs
5. webapps
6. temp
7. work

Start Tomcat

Open

Tomcat

bin

startup.bat

A command window appears.

Verify Tomcat

Open your browser.

Type:

`http://localhost:8080`

If the Tomcat homepage appears, the server is running successfully.

Stop Tomcat

Run:

shutdown.bat

inside the **bin** folder.

Step 3: Install Eclipse IDE

Download:

Eclipse IDE for Enterprise Java and Web Developers

Install Eclipse.

When Eclipse opens, choose a workspace.

Example:

`C:\Users\YourName\workspace`

Step 4: Configure Tomcat in Eclipse

Open Eclipse.

Go to

Window

Preferences

Server

Runtime Environments

Click

Add

Choose

Apache

Tomcat v10.1

Click

Next

Browse to your Tomcat installation folder.

Example

C:\Apache\Tomcat10

Click

Finish

Tomcat is now configured.

Step 5: Create a Dynamic Web Project

In Eclipse:

File

New

Dynamic Web Project

Project Name

ServletDemo

Target Runtime

Apache Tomcat

Click

Finish

Project Structure

Your project will look like:

ServletDemo

1. Java Resources

- 1.1 src

2. WebContent (or src/main/webapp)

- 2.1 index.html

- 2.2 WEB-INF

3. web.xml (optional when using annotations)

Step 6: Create Your First Servlet

Right-click

src

New

Servlet

Enter

Class Name

HelloServlet

Click

Finish

Eclipse generates the servlet class.

Step 7: Write the Servlet Code

Replace the generated code with:

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.*;

@WebServlet("/hello")
public class HelloServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request,
                          HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<h1>Hello World</h1>");
    }
}
```

Save the file.

Step 8: Run the Servlet

Right-click the project.

Choose

Run As

Run on Server

Select

Apache Tomcat

Click

Finish

Tomcat starts automatically.

Step 9: Open the Servlet in Browser

Open:

<http://localhost:8080/ServletDemo/hello>

Output:

Hello World

Execution Flow

User Opens Browser

1. Enter URL

2. HTTP Request
3. Tomcat Server
4. Servlet
5. doGet()
6. HTML Response
7. Browser Displays Output

Running example Programs

Suppose you create

```
@WebServlet("/date")
```

Then run:

```
http://localhost:8080/ServletDemo/date
```

Suppose you create

```
@WebServlet("/welcome")
```

Run:

```
http://localhost:8080/ServletDemo/welcome
```

Suppose your servlet reads parameters:

```
@WebServlet("/name")
```

Run:

```
http://localhost:8080/ServletDemo/name?user=Rahul
```


Output:

Welcome Rahul

Running HTML Form Programs

Create

index.html

Example

```
<form action="welcome">
```

Name

```
<input type="text" name="username">
```

```
<input type="submit">
```

```
</form>
```

Open

<http://localhost:8080/ServletDemo/index.html>

Enter:

Rahul

Click

Submit

Browser sends request

WelcomeServlet

Servlet displays

Welcome Rahul

Common Errors and Solutions

<i>Error</i>	<i>Cause</i>	<i>Solution</i>
404 Not Found	Incorrect URL or servlet mapping	Check the URL and @WebServlet path
Server Not Starting	Tomcat is not configured or port 8080 is already in use	Configure Tomcat correctly or change the port
ClassNotFoundException	Missing libraries or wrong package	Verify the project build path and imports
HTTP Status 500	Exception in servlet code	Check the Eclipse Console or Tomcat logs for the stack trace
Compilation Errors	Missing Servlet API or syntax errors	Ensure the project uses the correct Tomcat runtime and Servlet API

overview

1. Install JDK
2. Install Tomcat
3. Install Eclipse Enterprise
4. Configure Tomcat in Eclipse
5. Create Dynamic Web Project
6. Create Servlet
7. Write Servlet Code
8. Run on Server
9. Open Browser
10. Enter Servlet URL
11. View Output